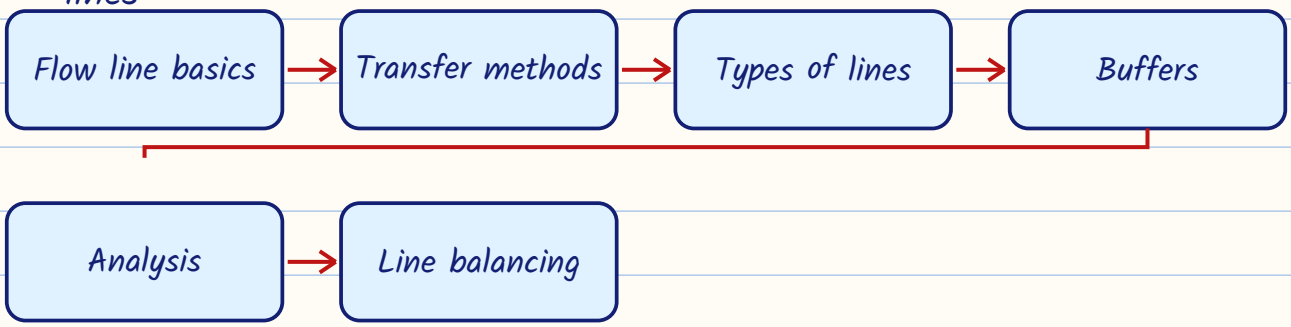
UNIT - II
Automated Flow Lines & Assembly Line Balancing

Automation and Robotics (23A03606) - handwritten study notes

Syllabus at a glance

- Automated flow lines, part transfer methods and mechanisms
- Types of flow lines, flow line with / without buffer storage
- Quantitative analysis of flow lines
- Assembly line balancing: assembly process and systems, assembly line
- Line balancing methods, ways of improving line balance, flexible assembly lines



1. Automated Flow Lines

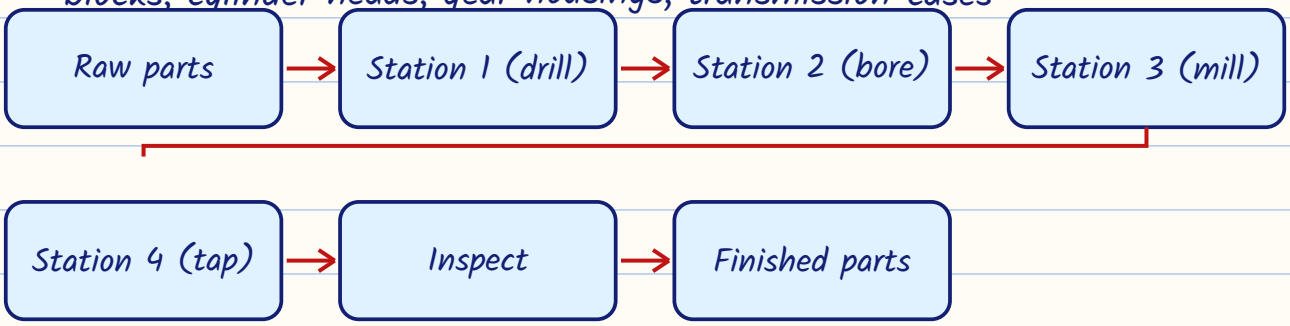
Definition

An **automated flow line** (transfer line) is a series of automatic workstations connected by a **work-part transfer system**. Raw work-parts enter at one end, are processed station by station and leave as finished (or semi-finished) parts at the other end.

- Each station does a **different operation** (drilling, boring, milling, tapping, inspection ...)
- All stations work **simultaneously** on different parts, so output = one part every cycle
- Part movement is **automatic** and stations are **synchronised** by a

common control system

- Used for **mass production** of identical or very similar parts: engine blocks, cylinder heads, gear housings, transmission cases



Disadvantages and Limitations

- Very high capital investment; economic only for large volumes
- Fixed (inflexible) layout: a change of product needs costly retooling
- **Line is only as strong as its weakest station**: one breakdown stops the entire line (no buffer)
- Needs skilled maintenance staff and spare tooling

Key idea: A flow line is a compromise between high production rate and loss of production caused by breakdowns of any one station.

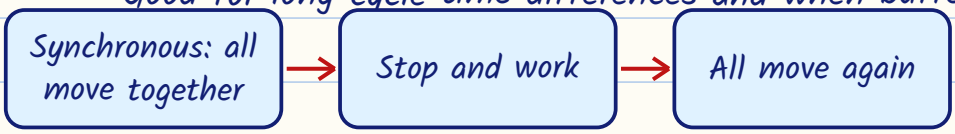
2. Part Transfer Methods and Mechanisms

Two Basic Methods

Part transfer system moves the work-parts from one station to the next. It also **locates and clamps** the part at the station. Two categories of motion exist:

- **Continuous transfer** - work-parts move continuously at constant speed along the line
  - Used in **non-synchronous / bottling / packaging / conveyor assembly** work
  - Workheads move with the part (tracking) or the operator works on the moving part

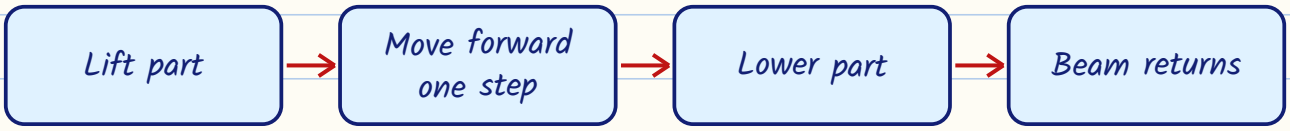
  - Simple, smooth, low shock loads
- **Intermittent (synchronous) transfer** - all parts move together, stop, work is done, then all move again
  - Used in **machining and automatic assembly lines**
  - Part is stationary while being processed, so precise location is possible
  - Motion time = transfer time T_r, then processing time
- **Asynchronous (power-and-free) transfer** - each part moves independently when its station is free
  - Parts can wait in small queues between stations; stations are not locked together
  - Good for long cycle time differences and when buffering is needed



Transfer Mechanisms - Linear Transfer

Used for in-line (straight) lines, parts move along the line.

- **Walking beam** - parts rest on fixed rails; a moving beam lifts, advances and lowers all parts one position. Very accurate, used for heavy parts.
- **Powered roller conveyor** - parts or pallets on driven rollers; simple and cheap.
- **Chain / belt conveyor** - parts move on a belt or chain, either continuously or in steps.
- **Overhead trolley / rail-guided carriers** - hanging parts, painting and plating lines.
- **Power-and-free** system - overhead rail with a power chain and a free trolley that can be released and stopped.
- **Cross-bar transfer** (shuttle bar) - a bar with fingers engages the pallet and pushes it forward; hydraulic or pneumatic drive.



Walking beam: parts are lifted, moved and set down exactly at the stations, so locating accuracy is very good.

Transfer Mechanisms - Rotary Transfer (Indexing)

Parts are carried on a circular table or dial that rotates in steps (indexes).

- **Rack and pinion** - linear reciprocating rack drives a pinion giving rotary motion; simple but limited speed.
- **Ratchet and pawl** - a pawl pushed by a cylinder advances the ratchet wheel by one tooth each stroke.
- **Rotary cam mechanism** - a barrel cam turns the table smoothly; accurate and fast.
- **Geneva mechanism** (Maltese cross) - a driving pin enters a slot in the driven wheel and turns it by one step while the crank turns, then the wheel locks and dwells.

Geneva Mechanism - Analysis

Let n = number of slots in the Geneva wheel. The driven wheel turns one slot angle of 360/n for each full rotation of the driving crank.

$$\text{Angle indexed per cycle} = 360 / n \text{ degrees}$$

$$\text{Crank angle during indexing} = 180 - 360 / n$$

$$T_r = (1/2 - 1/n) T_c$$

- Dwell fraction = 1 - Tr/Tc = 1/2 + 1/n; larger n gives longer dwell. Example: n = 6, Tc = 6 s, so Tr = (0.5 - 0.1667) x 6 = 2 s and 4 s dwell.
- Advantages: smooth start and stop, positive locking during dwell,

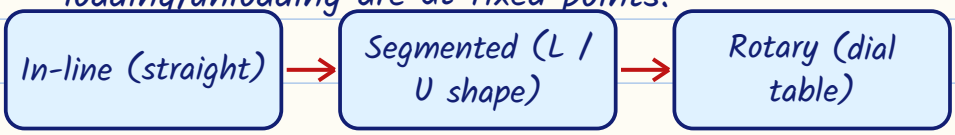
compact, reliable.

- Disadvantages: fixed index angle, jerk at pin entry for low n, limited to light and medium loads.

3. Types of Flow Lines

3.1 Based on Layout (Configuration)

- **In-line** - stations arranged in a straight line; the most common machining transfer line. Easy extension, good access for maintenance.
- **Segmented in-line** - two or more straight segments joined at right angles (L, U, rectangular). Saves floor space; operators can handle several segments.
- **Rotary** - stations around a circular dial table; work-parts are indexed. Compact and rigid, but the number of stations is limited and loading/unloading are at fixed points.



3.2 Based on Storage

- **Without buffer storage** - rigid line; all stations stop when any one stops
- **With buffer storage** - line is divided into **stages** by buffers; a failure in one stage does not stop the others immediately

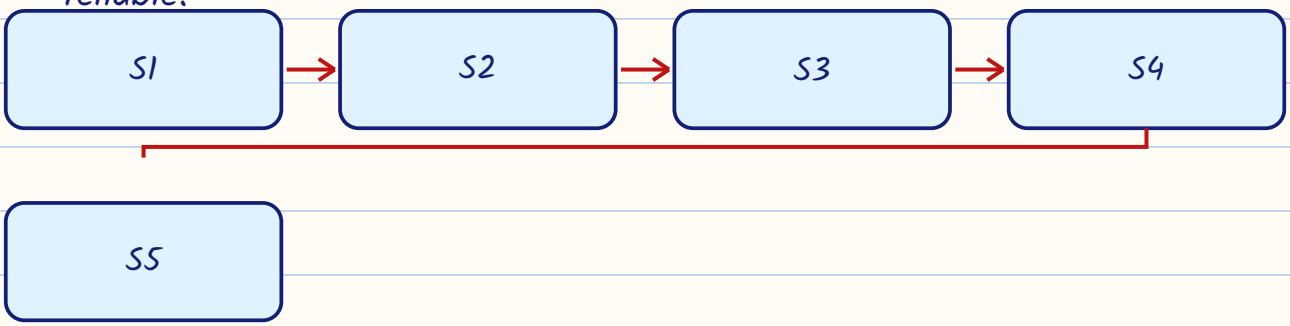
Choose in-line for long, large-volume machining lines; rotary for small parts and fewer operations.

4. Flow Line With and Without Buffer Storage

4.1 Without Buffer (Pure Series Line)

- All stations are linked in strict series; a breakdown at **any one** station stops all other stations.

- The line has the **lowest** **efficiency** but the lowest cost and simplest controls.
- Suitable when the number of stations is small and each station is highly reliable.

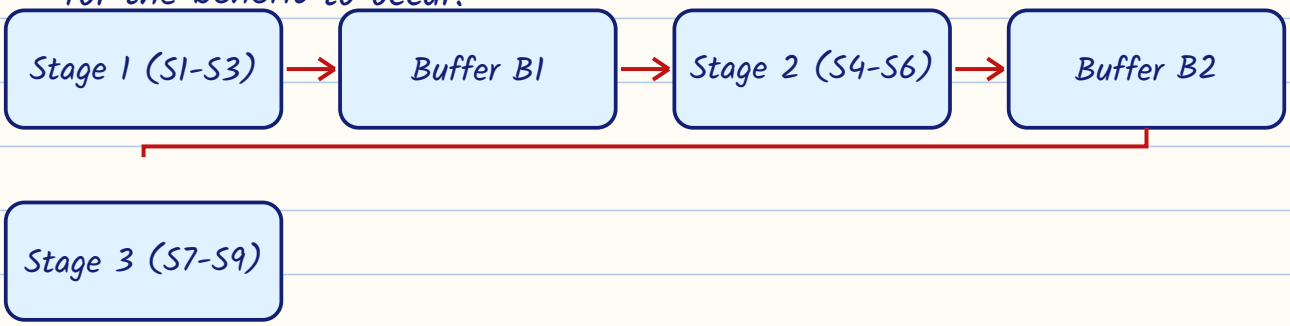


Failure of S3 means S1, S2, S4 and S5 are all forced to stop (no storage to absorb the stop).

4.2 With Buffer Storage

A **buffer** **storage** is a place between stages that stores work-parts temporarily so that the line downstream or upstream can keep running while one stage is down.

- When an upstream stage fails, downstream stages continue to work using parts from the buffer.
- When a downstream stage fails, upstream stages continue and fill up the buffer.
- Buffer must be **neither** **empty** **nor** **full** at the time of the breakdown for the benefit to occur.



Effects and Design Questions

- Larger buffer gives higher efficiency but extra **cost, floor space and WIP**
- Buffer locations: usually between stages of nearly equal reliability; sometimes before expensive or unreliable stations
- Efficiency improvement is **not linear**: first few buffer units give the largest gain; infinite buffer gives the **upper limit**

$$E_0 <= E_b <= E_{infinity}$$

5. Quantitative Analysis of Flow Lines

5.1 Notation

Symbol	Meaning
n	Number of workstations on the line
T_c	Ideal cycle time = max T_{si} + T_r (min)
T_{si}	Service time of station i, T_r = transfer time
F	Breakdown frequency of the line (stoppages per cycle)
T_d	Average downtime per breakdown (min)
T_p	Actual average production cycle time (min)
R_p	Production rate (pieces per min or hour)
E	Line efficiency = proportion of uptime
p_i	Probability that station i breaks down per cycle

5.2 Ideal Cycle Time

The line cycle time is decided by the **slowest station** plus the transfer time.

$$T_c = \max (T_{s1}, T_{s2}, \dots , T_{sn}) + T_r$$

- Ideal production rate (no breakdowns): $R_c = 1 / T_c$

5.3 Breakdowns of the Line

- Every station can stop due to tool failure, jam, adjustment, power failure,

etc.

- F = sum of station breakdown frequencies = $p_1 + p_2 + \dots + p_n$

$$F = \Sigma\, p_i \ (= n\,p \text{ , if all } p \text{ are equal})$$

- Downtime per cycle = $F\,T_d$

5.4 Average Production Cycle Time and Rate

$$T_p = T_c + F\,T_d$$

$$R_p = 1 / T_p = 1 / (T_c + F\,T_d)$$

5.5 Line Efficiency and Downtime Proportion

$$E = T_c / T_p = 1 / (1 + F\,T_d / T_c)$$

$$D = F\,T_d / T_p = 1 - E$$

- E is the fraction of time the line is productive; D is the fraction lost to breakdowns.
- Actual production rate in pieces per hour: $R_p = 60\,E / T_c$

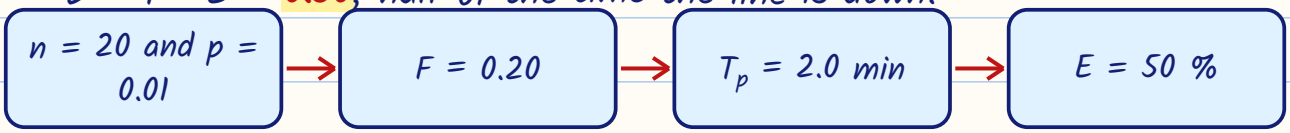
Remember: higher F or higher downtime lowers E. With n stations each with p, $F = n\,p$, so efficiency falls quickly as the number of stations grows.

Worked Example 1 (Series line without buffer)

A 20-station transfer line has $T_c = 1.0$ min and each station has breakdown probability $p = 0.01$ per cycle. Average downtime is $T_d = 5$ min. Find F, T_p, R_p, E and D.

- $F = n\,p = 20 \times 0.01 =$ **0.20 breakdowns/cycle**
- $T_p = T_c + F\,T_d = 1.0 + 0.2 \times 5 =$ **2.0 min**
- $R_p = 1 / 2.0 =$ **0.5 pieces/min = 30/hr**
- $E = T_c / T_p = 1.0 / 2.0 =$ **0.50 = 50 %**

- D = 1 - E = **0.50**; half of the time the line is down.



6. Analysis of Transfer Lines with Buffer Storage

6.1 Upper Bound Approach - Infinite Buffer

Assume a buffer of **infinite capacity** between every pair of stages, so one stage never has to stop because of another. Then each stage's efficiency is its own, and the line efficiency is limited only by the **worst stage**.

$$E_k = 1 / (1 + F_k T_d / T_c) \text{ for stage } k$$

$$E_{infinity} = \min (E_1, E_2, \dots)$$

6.2 Lower Bound - No Buffer

$$E_0 = 1 / (1 + (F_1 + F_2 + \dots) T_d / T_c)$$

6.3 Two-Stage Line with a Finite Buffer

- Stage 1 and stage 2 have frequencies F_1 and F_2, and a buffer of capacity b between them.
- Let $r = F_2 / F_1$; for equal downtime ($T_{d1} = T_{d2}$), the **probability that the buffer is neither empty nor full** is h(b).

$$E_b = E_0 + D_1' \, h(b) \, E_2$$

$$D_1' = F_1 T_d / (T_c + (F_1 + F_2) T_d)$$

$$h(b) = B/(B+1) + (T_c/T_d) \, L / ((B+1)(B+2)) \text{ for } r = 1$$

- Here $b = B (T_d / T_c) + L$, where B is an integer (0, 1, 2 ...) and L is the remainder.

- $h(0) = 0$ (no buffer, no gain) and $h(b)$ approaches 1 as b becomes very large.

Worked Example 4 (Two stages, buffer of 10)

Two equal stages: $F_1 = F_2 = 0.01$, $T_c = 1$ min, $T_d = 5$ min. Buffer capacity $b = 10$.

- $E_0 = 1 / (1 + 0.02 \times 5) =$ **0.909**
- $E_1 = E_2 = 1 / (1 + 0.01 \times 5) =$ **0.952**
- $D_1' = 0.05 / (1 + 0.02 \times 5) = 0.05 / 1.1 =$ **0.0455**
- $b = 10 = B \times 5 + L$ gives $B = 2$, $L = 0$, so $h(b) = 2/3 =$ **0.667**
- $E_b = 0.909 + 0.0455 \times 0.667 \times 0.952 =$ **0.938**

A buffer of only 10 parts recovers about two thirds of the possible gain.

- Place buffers where breakdown frequency is highest; divide the line into stages of roughly equal F; avoid too many buffers (cost, space, WIP).

7. Assembly Line Balancing

7.1 Assembly Process, Systems and Line

Assembly = joining components into a product (40-50 % of production cost). Systems: manual, automated, hybrid, synchronous, asynchronous and flexible assembly lines.

7.2 Assembly Line

An **assembly line** is a series of workstations; the product moves from station to station and workers (or machines) perform a fixed set of tasks at each. Tasks are called **work elements**.

- Work elements have time T_{ek} and **precedence constraints** (what must be done before what)
- Total work content: $T_{wc} = \Sigma\ T_{ek}$

7.3 Terms Used in Line Balancing

- Key terms: T_{wc} total work content, T_c cycle time, T_s station service time, delay time = $T_c - T_s$, precedence diagram, zoning constraint

7.4 Basic Relations

$$T_c = 60\ E / R_p \ (R_p \text{ in units per hour})$$

$$w_{min} = T_{wc} / T_c \ (\text{round up to the next integer})$$

$$E_b = T_{wc} / (w\ T_c) \ (\text{balance efficiency})$$

$$d = 1 - E_b = (w\ T_c - T_{wc}) / (w\ T_c) \ (\text{balance delay})$$

- Balance delay is the **idle time (loss)** because of imperfect balance.
- $E_b = 100\ \%$ only if every station is loaded to the full T_c.

Worked Example 5 (Cycle time and minimum stations)

A product has $T_{wc} = 28$ min. The required output is 40 units/hr and line efficiency is 0.95.

- $T_c = 60 \times 0.95 / 40 =$ **1.425 min**
- $w_{min} = 28 / 1.425 = 19.6$, so **20 workers** are required.

8. Line Balancing Methods

The goal: assign elements to stations to **minimise the number of stations** (or maximise E_b), obeying precedence and cycle-time limits.

8.1 Largest Candidate Rule (LCR)

- List all work elements in descending order of T_e.
- Start with station 1: choose the **largest** element that has no unassigned predecessors and **fits** in the remaining time.
- Repeat until no element fits; open the next station and continue until all elements are assigned.

8.2 Kilbridge and Wester Method (Column method)

- Draw precedence diagram and group elements in **columns** according to precedence (earliest possible column).
- Elements in column I come first, then II, and so on; within the column

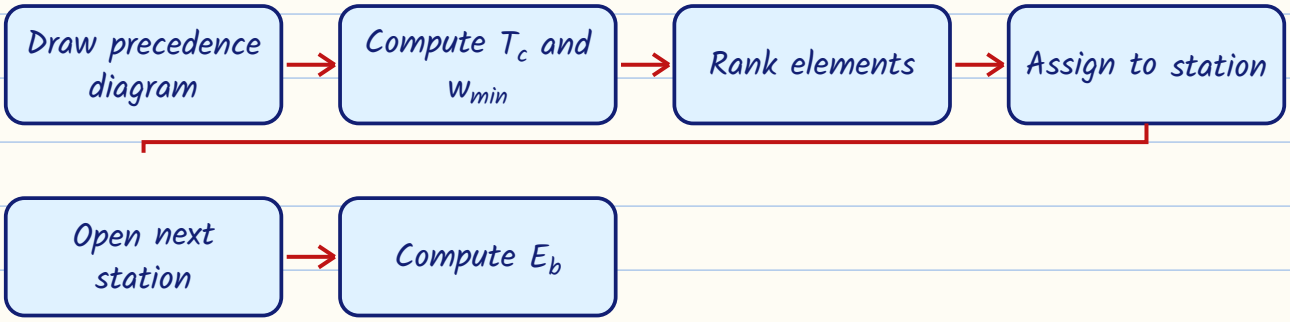
choose by longest time.

- Elements with a choice of column (free) are assigned to fill the gaps.
- Gives a better result than the LCR in many problems because it respects the **position** in the network.

8.3 Ranked Positional Weights (Helgeson and Birnie)

- For each element, **positional weight** RPW = its own time + the times of all elements that **follow** it in the precedence diagram.
- Rank the elements in descending RPW.
- Assign elements to stations in this order if precedence and T_c permit.

$$RPW_i = T_{ei} + \Sigma \text{ (times of all elements after } i \text{)}$$



Worked Example 6 (Largest Candidate Rule)

Elements (time in min, predecessors): 1 (0.5, none), 2 (0.3, after 1), 3 (0.8, after 1), 4 (0.2, after 2), 5 (0.6, after 3), 6 (0.4, after 4 and 5).

Required T_c = 1.0 min.

Element	Time	Predecessor
1	0.5	-
2	0.3	1
3	0.8	1
4	0.2	2
5	0.6	3
6	0.4	4, 5

- T_{wc} = 0.5 + 0.3 + 0.8 + 0.2 + 0.6 + 0.4 = **2.8 min**, w_{min} = 2.8 / 1.0 =

2.8, so **3 stations**

- Station 1: element 1 (0.5); candidates 2 and 3; 3 does not fit (1.3), so take 2 (0.8), then 4 (1.0). **S1 = {1, 2, 4}**, $T_{s1} = 1.0$
- Station 2: element 3 (0.8); element 5 would give 1.4, no fit. **S2 = {3}**, $T_{s2} = 0.8$
- Station 3: elements 5 (0.6) and 6 (0.4). **S3 = {5, 6}**, $T_{s3} = 1.0$

$$E_b = 2.8 / (3 \times 1.0) = 0.933 = 93.3\ \%$$

- Balance delay d = 6.7 %. Idle time exists only at station 2 (0.2 min).
- Same data by RPW: weights 1 = 2.8, 3 = 1.8, 5 = 1.0, 2 = 0.9, 4 = 0.6, 6 = 0.4 gives the same 3-station result (93.3 %).

9. Ways of Improving Line Balance

- **Divide long elements** into smaller ones so that they fill the gaps
- **Reassign** elements between neighbouring stations or **rearrange** precedence where possible
- **Use parallel workstations** for the bottleneck element (two stations doing the same task)
- Use **zoning of similar tasks** / multi-skilled operators who can move to help the busiest station
- Provide **mixed-model sequencing** to level the load across models
- **Reduce the cycle time** by using faster tools, jigs and fixtures, better workplace layout
- **Use underloaded stations** for additional tasks (inspection, packing)
- Use **buffer stock** between stations and **utility workers (floaters)** to cover absence
- Apply **work sampling and time study** to measure real element times, then re-balance periodically

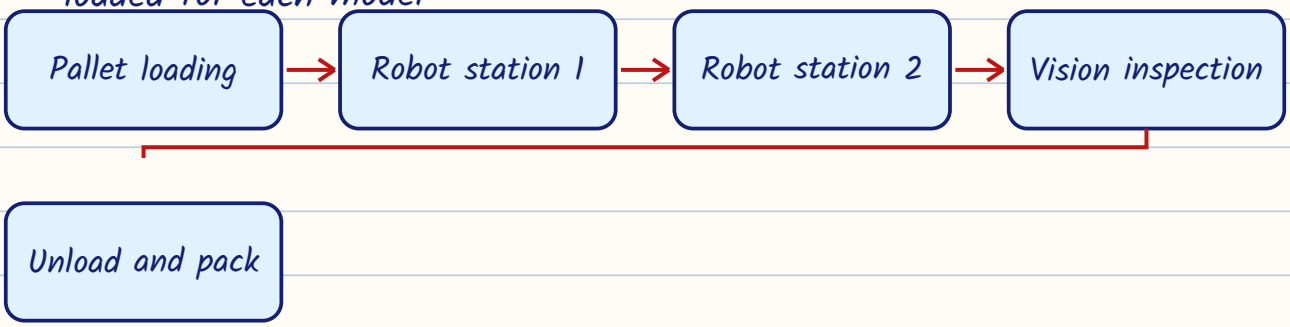
Remember: perfect balance (E_b = 100 %) is rare, so aim at **95 % or above** with minimum workstations.

10. Flexible Assembly Lines

Meaning

A **flexible assembly line (FAL)** consists of programmable assembly stations (robots, NC machines) linked by an automatic transport system, able to assemble **different products or variants** in small or mixed batches without major changeover.

- Stations: assembly robots (SCARA, Cartesian), pick-and-place units, screwdriving and vision inspection
- Transport: pallets on a conveyor, AGVs, shuttle systems with part identification (barcode, RFID)
- Control: central computer with **workstation controllers**; programmes loaded for each model



Features

- Flexibility for **mixed model** production and design changes
- Short setup, quick response to demand
- Higher quality and repeatability; reduced labour
- Integration with **CAD/CAM and ERP**
- Applications: electronic goods (PCBs, relays, motors), automobile parts, domestic appliances

11. Quick Revision Points

- Flow line = stations in series + automatic transfer; **synchronous** (stop-and-go), **asynchronous**, **continuous** transfer
- Mechanisms: walking beam, conveyor, cross-bar (linear); rack-pinion, ratchet-pawl, cam, **Geneva** (rotary)
- Layouts: in-line, segmented in-line, rotary; with or without buffers

$$T_c = \max T_{si} + T_r \mid T_p = T_c + F\,T_d \mid E = T_c / T_p$$

$$R_p = 1 / T_p \mid F = \Sigma\, p_i \mid D = 1 - E$$

- Buffer: $E_0 <= E_b <= E_{infinity}$; $h(b)$ = probability buffer neither empty nor full
- Line balancing: $T_c = 60\,E / R_p$; $w_{min} = T_{wc} / T_c$; $E_b = T_{wc} / (w\,T_c)$; $d = 1 - E_b$
- Methods: **LCR**, Kilbridge-Wester, Ranked positional weights**

Important Exam Questions

- 1. Define an automated flow line and explain its types with neat sketches. (5-10 marks)
- 2. Explain the part transfer mechanisms: walking beam, Geneva, rack and pinion, cam. (10 marks)
- 3. Differentiate a flow line with and without buffer storage; derive E for no buffer. (10 marks)
- 4. A 25-station line has $T_c = 0.8$ min, $p = 0.01$ and $T_d = 4$ min. Find F, T_p, R_p, E. (Answer: $F = 0.25$, $T_p = 1.8$, $R_p = 33.3$/hr, $E = 44.4$ %)
- 5. Explain the upper and lower bound analysis for a two-stage line with a buffer.
- 6. Explain line balancing methods with examples: LCR, Kilbridge-Wester, RPW. (10 marks)
- 7. How can the balance of an assembly line be improved? Write a note on flexible assembly lines.